

Lesson 3 · Sketching Gradient Functions

Learning intention

Sketch the graph of $f'(x)$ from the graph of $f(x)$, and sketch a possible $f(x)$ from the graph of $f'(x)$.

Success looks like

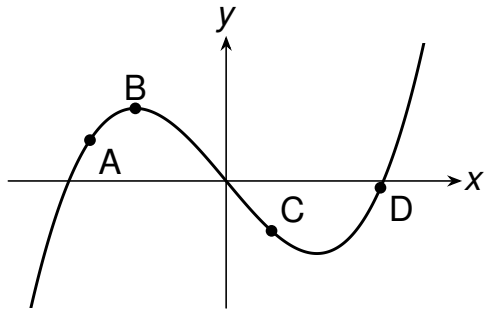
- I put the zeros of f' under the turning points of f .
- I put f' on the correct side of the axis in every interval.
- I can go backwards from f' to a possible f .

Do Now

1 Find the turning point of $y = x^2 - 6x + 1$ and state its nature.

2 For which x is $f(x) = x^3 - 3x^2$ decreasing?

3 Is the gradient at A, B, C and D positive, negative or zero?



The one idea

height of f' = steepness of f

graph of f

graph of f'

turning point

going up

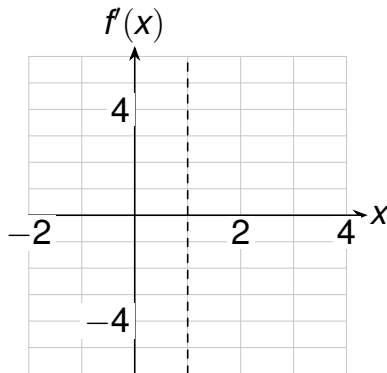
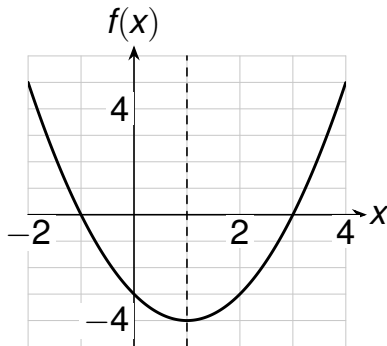
going down

steepest point

Lines and parabolas

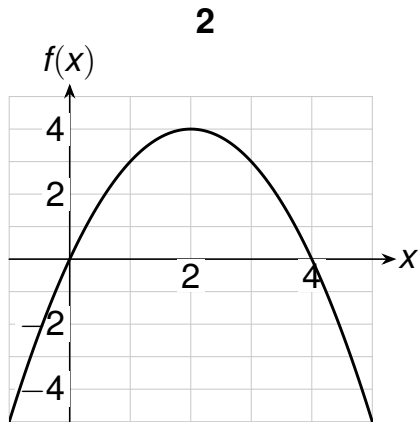
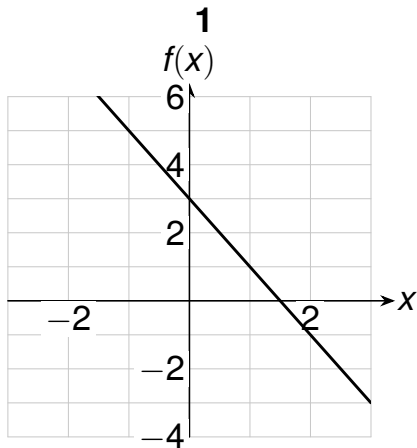
A straight line: f' is a **horizontal line**. A parabola: f' is a **straight line** through the x -value of its turning point.

1 Sketch $f'(x)$ for this parabola, on the axes on the right.



Practice

Sketch $f'(x)$ on the sketch page, grids 1 and 2.



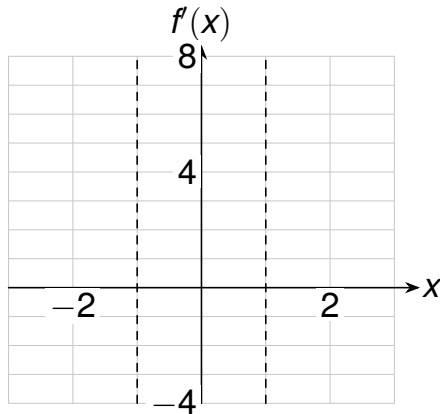
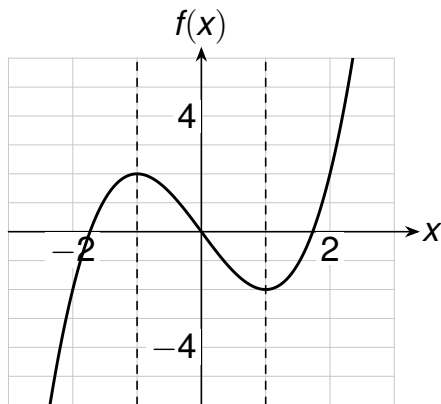
Cubics and quartics

Write it in this order

1. turning points of $f \rightarrow$ zeros of f' (use a ruler).
2. f going up $\rightarrow f'$ above the axis; going down $\rightarrow$ below.
3. Steepest point of $f \rightarrow$ turning point of f' .
4. Join smoothly. Check the type: cubic $\rightarrow$ parabola, quartic $\rightarrow$ cubic.

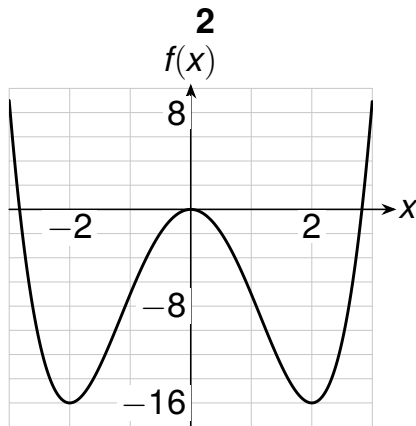
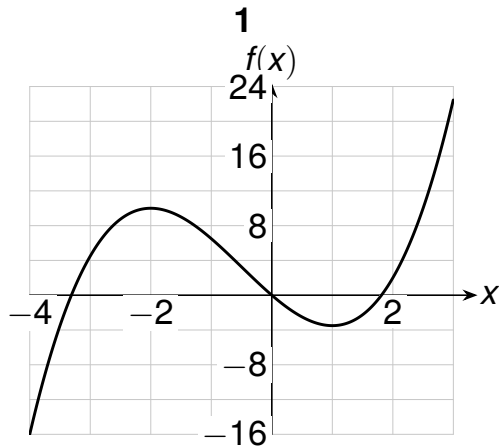
Worked example · cubic

2 Sketch $f'(x)$ for $f(x) = x^3 - 3x$, on the axes on the right.



Practice

Sketch $f'(x)$ on the sketch page, grids 3 and 4.



From f' back to f

f' graph

f graph

crosses from $+$ to $-$

crosses from $-$ to $+$

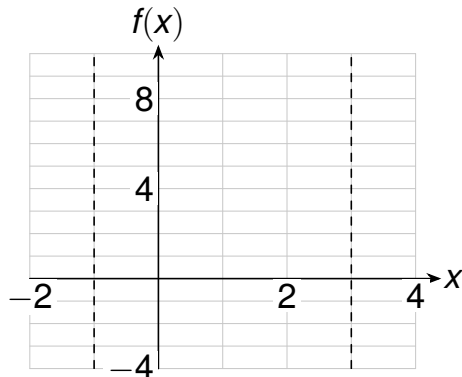
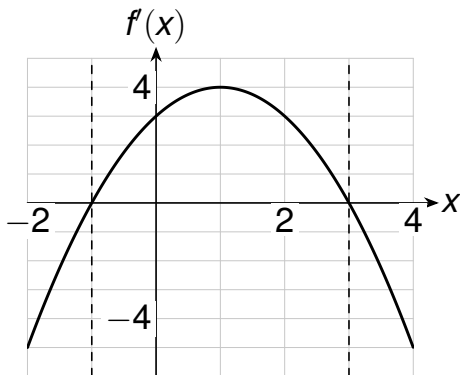
above the axis

below the axis

f' gives gradients only, so the height of f is unknown: sketch **a possible f** .

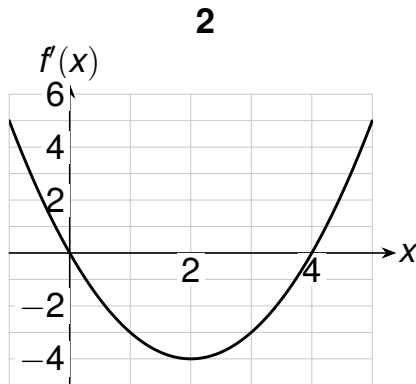
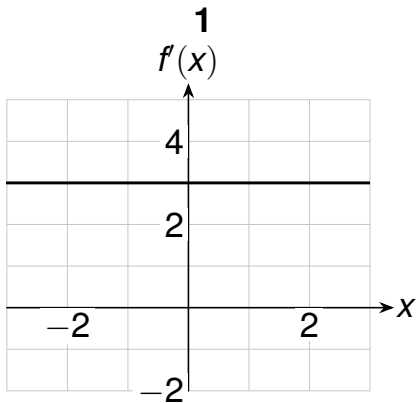
Worked example · backwards

3 Sketch a possible $f(x)$ for this $f'(x)$, on the axes on the right.



Practice

Sketch a possible $f(x)$ on the sketch page, grids 5 and 6.



Ways to lose marks

- L1 turning points of f become **zeros** of f' .
- L2 f going down $\rightarrow f'$ below the axis.
- L3 Parabola $\rightarrow$ line; cubic $\rightarrow$ parabola.
- L5 Zeros of f' are turning points of f .
- L6 Line up x -values with a ruler; label the graph.

Summary

1. Height of f' = steepness of f .
2. Turning points of $f \leftrightarrow$ zeros of f' .
3. Up: f' above the axis. Down: f' below.
4. Backwards: $+$ to $-$ is a maximum; the height of f is free.

Practice sheet · 30 minutes

Work alone on the practice sheet, from Question 1.

- Questions 1 and 2: warm-up, about 6 minutes.
- Questions 3 to 7: exam questions, about 24 minutes.
- Use a ruler to line up every key x -value.

Homework

Homework sheet: 14 questions, about 60 minutes

- Questions 1 to 11: Achieved and Merit sketches.
- Questions 12 and 13: Merit.
- Question 14: Extension (Excellence).

Optional extra: StudyTime Guide, “Sketching Gradient Functions” from page 19.